A finite-semigroup proof of the permutation-like matrix spectral assertion

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Abstract

A square matrix with one entry equal to one in each column and all other entries zero represents a map on a finite set. Finiteness makes two of its powers equal, proving that its nonzero eigenvalues are roots of unity. We give the complete argument and a Lean certificate, including the zero-eigenvalue characterization. This answers the assertion recorded as MathDB362146 and Conjecture 5.2 of Lippold's paper, without a claim of worldwide novelty or priority.

Theorem 1. *Let K be a field and I a finite index set. Let $A \in K^{I\times I}$ have exactly one entry equal to one in each column, with all other entries zero. Every eigenvalue $\mu \in K$ is zero or satisfies $\mu^k = 1$ for some integer $k > 0$. Moreover, zero is an eigenvalue if and only if A is not a permutation matrix.*

Proof. For each column j, let $f(j)$ be its unique unit-entry row. Then $A = T_f$, where $(T_f)_{ij} = 1$ if $i = f(j)$ and zero otherwise. Multiplication gives $T_f T_g = T_{f\circ g}$, so $T_f^n = T_{f^n}$. There are finitely many functions $I \to I$. Hence $f^p = f^q$ for some integers $0 \le p < q$, and $A^p = A^q$.

For an eigenvector $v \neq 0$ with $Av = \mu v$, induction gives $A^n v = \mu^n v$. Thus $\mu^p v = \mu^q v$. A nonzero coordinate of v implies $\mu^p = \mu^q$. If $\mu \neq 0$, division gives $\mu^{q-p} = 1$, proving the first assertion.

If f is not injective, choose $i \neq j$ with $f(i) = f(j)$. Then $T_f(e_i - e_j) = e_{f(i)} - e_{f(j)} = 0$, while $e_i - e_j \neq 0$. Thus zero is an eigenvalue. If f is injective, finiteness makes it bijective, and $T_{f^{-1}} T_f = I$ rules out a nonzero kernel vector. Finally, f is injective exactly when each row of T_f, as well as each column, has one unit entry; this is the permutation-matrix condition. □

Relationship to the source. Lippold [1] defines these matrices in Definition 2.1 and proves their multiplicative closure in Proposition 2.1. Corollary 4.5 gives the spectral assertion for dimensions two and three, Proposition 4.2 treats pre-row matrices, and Proposition 4.4 and Corollary 4.7 give the zero-eigenvalue and invertibility statements. Those are credited antecedents. The finite-set argument above removes the dimension and pre-row restrictions. The paper itself notes after Conjecture 5.2 that the zero-eigenvalue part is already settled. The identity is a permutation matrix, so a separate exclusion of the identity in the conjecture is redundant.

Formal coverage and limitations. `PermutationSpectrum.lean` proves the full theorem for every field and finite index type. It constructs the finite-map representation, proves multiplication and powers, obtains repeated powers by the pigeonhole principle, and constructs the zero eigenvector or inverse explicitly. The final statement uses mathlib's `Module.End.HasEigenvalue`; a separately proved equivalence connects it with the nonzero-vector equation. The declaration `exact_mathdb_conclusion` also states the redundant identity exclusion. No classification of Jordan blocks or algebraic multiplicities is claimed.

The build uses Lean 4.34.1 and mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

It prints every local type and its kernel axioms, and permits only standard axioms. The compiler and precompiled dependency objects remain trusted. A bounded source search found no separate general solution; this is not an exhaustive literature review. Original prose is CC BY 4.0 and original code MIT. No external review or submission is asserted.

References

[1] S. R. Lippold, *Permutation-Like Matrices*, arXiv:2403.02478v1, 2024. https://arxiv.org/html/2403.02478v1.

[2] MathDB362146, *The eigenvalue conjecture for permutation-like matrices*, retrieved October 25, 2026. https://mathdb.com/p/362146.